

Returning the Determinant-Two Möbius Corridor: Terminal-Window Loss, Atomic Prefixes, and Split Log/Power Ledgers

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Abstract

TPC-149 proves power-of-log cancellation for the actual determinant-two Möbius periodic core at all scales outside a local exceptional set. We determine exactly what is lost when this theorem is restricted to a terminal window and then sampled by a physical prefix rule. If the source exceptional exponent is κ_{src} and $\log \omega = (\log X)^{\theta+o(1)}$, then its normalized terminal-window restriction directly certifies only

$$\kappa_{\text{exc}} = \kappa_{\text{src}} + \theta - 1.$$

A positive source density saving may therefore disappear on a short window.

We also give a sharp deterministic-prefix firewall. Adding any finite list of requested endpoints to an exceptional set does not change its logarithmic measure. Hence exceptional density alone cannot certify a prescribed discrete prefix family. A pointwise theorem, proved exceptional avoidance, or a quantitative smoothing and selector crosswalk is required.

Finally, we separate the power-of-log return ledger from the fixed- X -power ledger. Even a positive logarithmic exponent has $\sigma_{\text{power}} = 0$, so it cannot pay a positive physical endpoint loss or $1/400$. Because the frontier occurrence lift is still missing, the current actual return is NOT-TESTABLE, not positive L2 and not a prime-pair result.

1 The sourced core and the missing occurrence lift

TPC-149 proves that there are constants $\eta_0, \kappa_{\text{src}} > 0$ and, for every ambient X , a set

$$\mathcal{E}_X^* \subset [\sqrt{X}, X]$$

such that

$$\frac{1}{\log X} \int_{\mathcal{E}_X^*} \frac{dt}{t} \ll (\log X)^{-\kappa_{\text{src}}}, \quad (1)$$

and the literal determinant-two Möbius periodic core has power-of-log cancellation at each $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X^*$ [1, 4].

This core theorem is uniform under a small-polylogarithmic bound on the product of the progression modulus and exact periodic period. It does not include a nonperiodic physical multiplier, a generic phase, an arbitrary interval or every prefix. In particular, it is not the four-sign Fejér input required by TPC-130 [2].

Moreover, the current TPC-143–146 frontier interface has not yet constructed the occurrence records carrying

$(a, s, d, u), \quad as, \quad \text{residue}, \quad \text{ordered interval}, \quad R, \quad \text{masks, prefixes and coefficient mass}.$

These fields are REQUIRED_MISSING. Thus

$$\boxed{\text{H1.frontier_occurrence_lift} = \text{NOT-TESTABLE.}} \quad (2)$$

The results below give the legal return once those records exist; they do not manufacture them.

2 A local exceptional set on a terminal window

Let

$$J_X = [X/\omega, X], \quad m_{J_X}(E) = \frac{1}{\log \omega} \int_{E \cap J_X} \frac{dt}{t}.$$

We first use one ambient shell, so we explicitly require

$$J_X \subset [\sqrt{X}, X], \quad \text{equivalently} \quad 1 < \omega \leq \sqrt{X}. \quad (3)$$

A longer window needs a declared shell decomposition and the union of all shell-level exceptional sets.

Proposition 2.1 (Local shell to terminal window). *Under (1) and (3),*

$$m_{J_X}(\mathcal{E}_X^*) \ll \min \left\{ 1, \frac{(\log X)^{1-\kappa_{\text{src}}}}{\log \omega} \right\}. \quad (4)$$

If

$$\log \omega = (\log X)^{\theta+o(1)}, \quad (5)$$

then the exponent supplied by this estimate is

$$\kappa_{\text{exc}} = \kappa_{\text{src}} + \theta - 1. \quad (6)$$

It is positive only if $\kappa_{\text{src}} + \theta > 1$.

Proof. The numerator defining m_{J_X} is at most the integral over $\mathcal{E}_X^*$, which is $O((\log X)^{1-\kappa_{\text{src}}})$ by (1). Divide by $\log \omega$, and also use $m_{J_X} \leq 1$. Substitution of (5) gives (6). $\square$

Remark 2.2 (No duplicated pair census). TPC-149 already forms one exceptional set by unioning over all allowed (a, s) in its universal envelope. Its pair census must not be charged again in (4). A record-aware variant may instead union only the actually occurring pairs and endpoint dilates, but then those counts must be entered exactly once.

3 A sharp deterministic-prefix firewall

Theorem 3.1 (Discrete endpoints are invisible to density). *Let $T_X \subset [\sqrt{X}, X]$ be any finite set of deterministic scales. For every measurable $E_X \subset [\sqrt{X}, X]$,*

$$\int_{E_X \cup T_X} \frac{dt}{t} = \int_{E_X} \frac{dt}{t}.$$

Consequently, an exceptional-set density theorem alone cannot imply that even one prescribed endpoint in T_X lies outside the exceptional set.

Proof. Every finite set has Lebesgue, hence logarithmic, measure zero. Therefore adjoining T_X leaves the integral unchanged. Both E_X and $E_X \cup T_X$ obey exactly the same density bound, while the latter contains every requested endpoint. $\square$

This is a logical nonimplication, not a claim that the arithmetic exceptional set actually contains the endpoints. It shows that density information cannot decide the question.

Let ν_X be the probability measure with which a proposed physical synthesis samples J_X . TPC-140 uses the domination condition

$$\nu_X(A) \leq K_X m_{J_X}(A) \quad (7)$$

for every Borel set A [3].

Corollary 3.2 (Atomic selector stop). *If ν_X gives positive mass to any deterministic endpoint, no finite K_X satisfies (7).*

Proof. Apply (7) to the singleton endpoint. Its m_{J_X} -measure is zero, while its ν_X -mass is positive. $\square$

There are three legal exits:

- (i) a pointwise theorem uniform in every actual prefix;
- (ii) an independent theorem proving that the requested endpoints avoid the actual exceptional sets; or
- (iii) an exact smoothing identity, a quantitative stability bound and selector domination for the smoothed measure.

4 The power-of-log return ledger

Suppose all occurrence records and normalizations have been supplied. Write

$$\begin{aligned}\varepsilon_{\text{corr}} &\ll (\log X)^{-\kappa_{\text{corr}}+o(1)}, \\ m_{J_X}(\mathcal{E}_{\text{tot}}) &\ll (\log X)^{-\kappa_{\text{exc}}+o(1)}.\end{aligned}\tag{8}$$

Boundedness and the good/bad split give

$$\kappa_{\text{aff}} = \min\{\kappa_{\text{corr}}, \kappa_{\text{exc}}\}.\tag{9}$$

Let the ratio of the literal outer absolute reassembly mass to the declared physical comparison mass cost $(\log X)^{\kappa_{\text{cen}}+o(1)}$. Let selector and nonperiodic weight/Abel return cost respectively $(\log X)^{\kappa_{\text{sel}}+o(1)}$ and $(\log X)^{\kappa_{\text{BV}}+o(1)}$. Unknown costs are not zero. If a separate remaining tail saves $(\log X)^{-\kappa_{\text{tail}}+o(1)}$, then:

Theorem 4.1 (Typed logarithmic return). *Under the preceding proved inputs, the exponent certified by their triangle-inequality return is*

$$\boxed{\kappa_{\text{returned}}^{\text{cert}} = \min\{\kappa_{\text{tail}}, \kappa_{\text{aff}} - \kappa_{\text{cen}} - \kappa_{\text{sel}} - \kappa_{\text{BV}}\}}.\tag{10}$$

For the exact quotient-Möbius core of TPC-149 there is no squarefree tail, so the first entry is omitted unless another literal remainder is present.

Proof. On good scales, sum the component estimates against the actual outer absolute mass. On bad scales, use boundedness and their normalized mass. This gives (9); the three multiplicative return costs subtract their exponents. A separately bounded tail is added, so the weaker of the two savings survives. $\square$

Periodic residue classes inside one TPC-149 record have already been reassembled with their native densities and do not contribute to κ_{cen} . Distinct outer records and the mismatch between their arithmetic and physical comparison masses may do so. The displayed exponent is a guaranteed ledger output, not a claim that the true error has exactly this order; it gives decay only when it is strictly positive.

5 The separate fixed- X -power ledger

Proposition 5.1 (Logarithmic saving has zero X -power). *For every fixed $\kappa > 0$, the largest fixed $\sigma \geq 0$ certified by*

$$(\log X)^{-\kappa+o(1)} \leq X^{-\sigma+o(1)}$$

is $\sigma = 0$.

Proof. For every fixed $\sigma > 0$,

$$X^\sigma (\log X)^{-\kappa} \longrightarrow \infty.$$

Thus a power of logarithm decays more slowly than every positive power of X . $\square$

Consequently,

$$\boxed{\sigma_{\text{power}} = 0} \tag{11}$$

for the present corridor, even when $\kappa_{\text{returned}}^{\text{cert}} > 0$. It cannot pay a positive physical occurrence loss, the local 1/400 target, or a strict endpoint. This is a scale comparison, not a theorem that a future stronger arithmetic estimate is impossible.

6 Actual status and machine contract

The deterministic audit imports TPC-147–149 by content hashes for integrity only, records every occurrence field required from TPC-143–146, and preserves (2). Its rational ledger contains a hypothetical positive log case and an equality case that must stop. Neither is evidence about the actual packet. The actual record retains unknown values and is NOT-TESTABLE.

Statement	Level and status
Terminal-window estimate (4)	Exact measure theorem, PROVED.
Deterministic-prefix nonimplication	Exact firewall, PROVED.
Logarithmic ledger (10)	Conditional typed L1.
TPC-149 actual Möbius periodic core	L1, PROVED.
Frontier occurrence lift	NOT-TESTABLE, first structural missing item.
Physical weight, generic phase, all-prefix return	OPEN after occurrence lift.
Four-point H3 or positive L2	Not proved.
Positive X -power, 1/400, endpoint pass	Not proved.
Prime-pair or twin-prime theorem	Not proved.

Thus the new corridor is mathematically real but remains upstream of the decisive physical and all-prefix gates. Its correct output is a stronger L1 arithmetic node plus an explicit return obstruction, not an H3 or endpoint certificate.

References

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